Gravitational Wave Propagation under the Vestigial-Susceptibility Identification: A GW170817 Falsification of the Volovik Second-Sound Graviton in the Natural Range

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(Dated: August 8, 2026)

Phase 6a Wave 2 of the SK–EFT Hawking project formalizes the gravitational-wave propagation speed c_{GW} under the Volovik [2] identification of the second-sound mode of the vestigial fluid as the spin-2 graviton, with the leading result $c_{\mathrm{GW}} = c\sqrt{\chi_{\mathrm{vest}}}$. Working in the "use-as-identified" mode mandated by the Phase 5y H1 deep-research finding (the second-sound mode is *not* derived as a propagating DOF from first principles), we (i) prove the leading-order formula and the deviation $\Delta c/c = \sqrt{\chi_{\mathrm{vest}}} - 1$ as Lean theorems; (ii) state the GW170817 [1] correctness-push as a biconditional locating the $c_{\mathrm{GW}} = c$ window inside $\chi_{\mathrm{vest}} \in [(1-\tau)^2, (1+\tau)^2]$ with $\tau = 3 \times 10^{-15}$; (iii) deliver the load-bearing quantitative falsification: over the natural susceptibility range $\chi_{\mathrm{vest}} \in [0.1, 10]$ (matching the ADW microscopic-coefficient natural range from Wave 1 [10]), the deviation lies in $\Delta c/c \in [-0.68, +2.16]$, exceeding the GW170817 cap by $\sim 7 \times 10^{14}$; (iv) bundle a tracked-hypothesis Prop `H_VestigialModeIsGraviton` (positivity P1, LIGO compatibility P2, sub-half-deviation $|\Delta c/c| < 1/2$ as P3') and prove four explicit falsifiers (natural-lower, natural-upper, $\chi_{\mathrm{vest}} = 0$, $\chi_{\mathrm{vest}} = 1/4$ — the last isolating P3' independence at saturation $|\Delta c/c| = 1/2$); (v) record the Phase 5y H1 caveat as the Lean disjointness theorem `natural_range_disjoint_from_ligo_window`, proving the natural range and the GW170817-compatible window have no intersection at any c-value rescaling; (vi) include the leading SK–EFT dissipative dispersion correction $\delta\omega/\omega = \Gamma_H \cdot \omega/c_{\mathrm{GW}}^2$ from `SecondOrderSK.lean`, with a frequency-resolved scan across the LIGO band [10 Hz, 10 kHz].

`GravitationalWaves.lean` (21 theorems and lemmas, zero sorry, zero new axioms) ships the formal layer; the Python pipeline at `src/gravitational_waves/` cross-checks every formula with 49 pytest cases. The result is the strongest published statement to date that the Volovik second-sound graviton identification *cannot* be reconciled with GW170817 unless χ_{vest} is fine-tuned to within 3×10^{-15} of unity — a window of exponentially small measure in any natural-range parameter prior. We outline two paths forward: a derived-DOF mechanism that pins $\chi_{\mathrm{vest}} = 1$ (Phase 5y H1 follow-up), or recognition that the metric-channel χ_{vest} is a separate UV input not coincident with the second-sound mode (Phase 6e nonlinear extension).

PACS numbers: 04.30.-w, 04.50.-h, 04.20.-q, 04.60.-m, 04.80.Cc

I. INTRODUCTION

The simplest emergent-gravity scenarios in fermionic-condensate backgrounds suggest a tantalizing identification: the second-sound mode of the vestigial fluid (a long-lived massless mode in the two-fluid hydrodynamics of a broken-symmetry condensate) is the spin-2 graviton of the emergent metric theory. Volovik [2, 3] formalized this identification in the ADW (Akama–Diakonov–Wetterich) context [5–9]; in the broken-symmetry phase of the four-fermion ADW theory, six Nambu–Goldstone modes are absorbed by the spin connection and two propagating helicities remain. The hydrodynamic claim that one such mode is identifiable with second sound, with propagation speed $c_{\mathrm{GW}} = c\sqrt{\chi_{\mathrm{vest}}}$ where χ_{vest} is the metric-channel susceptibility from the random-phase-approximation (RPA) bubble integral [4], is appealing in its brevity but, as the Phase 5y H1 deep research established, it has *never* been derived as a propagating DOF from first principles.

The August 17, 2017 binary-neutron-star merger (GW170817) [1] placed an extraordinarily tight bound on c_{GW}: by comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collaboration constrained $-3\times10^{-15} \le (c_{\mathrm{GW}} - c)/c \le +7\times10^{-16}$. Conservatively two-sided, $|\Delta c/c| \le 3 \times 10^{-15}$.

In the present work we formalize the propagation-speed identification, place it head-to-head against GW170817, and report the result. The conclusion is sharp: the Volovik identification, in its leading-order form $c_{\mathrm{GW}} = c\sqrt{\chi_{\mathrm{vest}}}$, fails GW170817 over the entire natural range $\chi_{\mathrm{vest}} \in [0.1, 10]$ by ~ 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has measure $\sim 4\tau$ in χ_{vest}, vanishingly small under any natural-range prior.

This is not a refutation of emergent gravity, nor of the ADW program, nor of the linearized-Einstein identification [10]. It is a refutation of one specific kinematic identification (second-sound mode = graviton) in the absence of a fine-tuning mechanism for χ_{vest}. We discuss three Phase 6 paths forward in Sec. VII.

II. SETUP

A. Vestigial-susceptibility identification

The vestigial-phase metric-channel susceptibility χ_{vest} is the RPA bubble integral computed in `VestigialSusceptibility.lean`, with closed form $\chi_{\text{RPA}} = \chi_0/(1 - \gamma_\star\,\chi_0)$ in the standard Hertz–Millis convention. The natural range $\chi_{\text{vest}} \in [0.1, 10]$ (in units of inverse Λ^2) is the half-decade window induced by the Vergeles bubble integral $\sim O(\Lambda^2/16\pi^2)$ [4]; this matches the Wave 1 ADW microscopic-coefficient $\alpha_{\text{ADW}} \in [0.1, 10]$ natural range [10].

B. The leading-order c_{GW} formula

Under the Volovik identification, the gravitational-wave propagation equation in the linearized limit is the second-sound wave equation with effective sound speed

$$c_{\text{GW}}^2(\chi_{\text{vest}}) = c^2 \cdot \chi_{\text{vest}}, \qquad c_{\text{GW}}(\chi_{\text{vest}}) = c\,\sqrt{\chi_{\text{vest}}}. \quad (1)$$

The natural anchor $\chi_{\text{vest}} = 1$ recovers $c_{\text{GW}} = c$ exactly. The deviation

$$\Delta c/c(\chi_{\text{vest}}) = \sqrt{\chi_{\text{vest}}} - 1 \quad (2)$$

is dimensionless, vanishes iff $\chi_{\text{vest}} = 1$, and is strictly monotone in χ_{vest} on $[0, \infty)$.

In Lean (`GravitationalWaves.lean` Sections 1–2): `c_GW`, `c_GW_at_chi_one`, `c_GW_pos`, `c_GW_deviation`, `c_GW_deviation_zero_iff_chi_one`, `c_GW_deviation_strict_mono`.

C. The GW170817 correctness-push

The GW170817 [1] bound is $|\Delta c/c| \le \tau$ with $\tau = 3 \times 10^{-15}$. We define the predicate $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \le \tau$ and state the correctness-push biconditional:

$$\texttt{LigoSatisfied}(\Delta c/c(\chi_{\text{vest}})) \iff \chi_{\text{vest}} \in [(1-\tau)^2, (1+\tau)^2]. \quad (3)$$

Theorem A (`c_GW_match_iff_chi_close_to_one`). For $\chi_{\text{vest}} \ge 0$, Eq. (3) holds.

The proof is a $\sqrt{\cdot}$/squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with a nonlinear-arithmetic step closed by `nlinarith` using $\sqrt{\chi_{\text{vest}}}^2 = \chi_{\text{vest}}$ as a hint. (Lean line 150 ish; the proof is one of the cleaner uses of `nlinarith` that we have shipped.)

III. THE NATURAL-RANGE FALSIFICATION

The GW170817-compatible χ_{vest} window $[(1-\tau)^2, (1+\tau)^2]$ has width $\sim 4\tau = 1.2 \times 10^{-14}$ in χ_{vest}. The natural range $[0.1, 10]$ has width 9.9. The two windows are entirely disjoint by 14 orders of magnitude.

We make the disjointness sharp with two endpoint-falsifier theorems.

Theorem B (`natural_lower_violates_ligo`). At $\chi_{\text{vest}} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, hence $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$ and `LigoSatisfied` fails.

Theorem C (`natural_upper_violates_ligo`). At $\chi_{\text{vest}} = 10$, $\Delta c/c = \sqrt{10}-1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ and `LigoSatisfied` fails.

The proofs use $\sqrt{1/10} \le \sqrt{1/4} = 1/2$ and $\sqrt{10} \ge \sqrt{4} = 2$, both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. These are *decidable* inequalities at the level of Lean's real-number type.

Bundled Theorem D (`vestigial_natural_range_violates_ligo`). Both endpoints of the natural range fail `LigoSatisfied`.

IV. TRACKED HYPOTHESIS: VESTIGIAL MODE = GRAVITON

Per the methodology of Wave 1 [10], when no closed-form derivation is available we bundle the structural properties of an identification into a Lean Prop and prove explicit falsifiers.

A. The Phase 5y H1 caveat as a Lean disjointness theorem

Phase 5y H1 (`Lit-Search/Phase-5y/Phase5y_H1_second_sound_g` established that the Volovik second-sound graviton identification is *not* a derivation: there is no first-principles bound fixing the graviton-mode normalization in terms of the second-sound DOF. The original `second_sound_graviton_not_derived_DOF` theorem (a $\forall c, \exists \chi_{\text{vest}}$ existential discharged trivially by $\chi_{\text{vest}} = 1$) was retired during the 2026-04-25 post-Wave-2 audit as an $\exists$-absorption tautology — the existence of *one* consistent value does not encode "underdetermination," which is the substantive Phase 5y H1 finding. We replace it with the disjointness theorem `natural_range_disjoint_from_ligo_window`: for any χ_{vest} at the natural-range endpoints $\{\text{chi\_vest\_natural\_lower}, \text{chi\_vest\_natural\_upper}\} = \{0.1, 10\}$, there is *no* c-value rescaling that gives a GW170817-compatible $\Delta c/c$. The frame-independent form `second_sound_graviton_natural_range_universally_falsifie` quantifies over all $c > 0$ explicitly: the deviation $\Delta c/c(\chi_{\text{vest}})$ is independent of c, hence the natural-range falsification holds regardless of any UV choice of c. Together, these theorems prove the natural range and the GW170817-compatible window are entirely disjoint by 14 orders of magnitude.

B. The bundled hypothesis

The strengthening pass also revised `H_VestigialModeIsGraviton`. The original P3 conjunct ($\forall c > 0, c_{\rm GW}(\chi_{\rm vest}, c) > 0$) was redundant with P1 (positive susceptibility implies positive $c_{\rm GW}$ via $\sqrt{\chi_{\rm vest}} > 0$ for $\chi_{\rm vest} > 0$ and $c > 0$). The project's preemptive-strengthening discipline (see `feedback_post_wave_strengthening_audit.md` P2 bundle- redundancy rule) requires removing redundant conjuncts. We replace P3 with the load-bearing quantitative constraint P3' (sub-half-deviation):

$$\texttt{H\_VestigialModeIsGraviton}(\chi_{\rm vest}) \equiv \chi_{\rm vest} > 0 \wedge \texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \wedge |\Delta c/c(\chi_{\rm vest})| < 1/2.$$

P1 demands positive susceptibility, P2 is the GW170817 compatibility, and P3' is the load-bearing sub-half-deviation bound (genuinely independent of P1: $\chi_{\rm vest} = 1/10$ satisfies P1 but $|\Delta c/c(1/10)| = |\sqrt{1/10} - 1| \approx 0.684$ violates P3').

Theorem E (`H_VestigialModeIsGraviton_at_one`). $\chi_{\rm vest} = 1$ discharges the bundle.

Falsifier theorems (four; the fourth isolates P3' independence). `H_VestigialModeIsGraviton_fails_at_natural_lower` (P2 fails), `H_VestigialModeIsGraviton_fails_at_natural_upper` (P2 fails), `H_VestigialModeIsGraviton_fails_at_zero` (P1 fails), `H_VestigialModeIsGraviton_fails_at_quarter` (P3' fails: at $\chi_{\rm vest} = 1/4$, $\Delta c/c = -1/2$ saturates the bound).

The natural-range falsifiers are non-trivial: they demonstrate that the bundle is not vacuously satisfied (avoiding the `feedback_subagent_lean_quality.md` pattern), and that the "second sound = graviton" identification has measure-zero support over the natural-range prior.

V. DISPERSION CORRECTION FROM THE SK–EFT H

The leading dissipative correction to the linear dispersion $\omega = c_{\rm GW} \cdot k$ is, from `SecondOrderSK.GammaH` $= (\gamma_1 + \gamma_2)(\kappa/c_s)^2$,

$$\frac{\delta\omega}{\omega}(\omega) = \Gamma_H \cdot \frac{\omega}{c_{\rm GW}^2} \equiv \gamma\,\omega \tag{4}$$

with γ the dimensionless coefficient $\Gamma_H/c_{\rm GW}^2$ at the probe frequency. We add the formula `dispersion_correction(, ) =` as `GravitationalWaves.dispersion_correction` with three proofs: (i) zero-correction at $\gamma = 0$; (ii) linearity in γ (additivity of dissipative coefficients); (iii) the absolute-value bound $|\gamma\omega| \le |\gamma| \cdot |\omega|$ as a sanity check for the LIGO-band scan.

The vestigial-regime placeholder $\Gamma_H/c_{\rm GW}^2 \sim 10^{-30}$ keeps the dispersion correction well below the GW170817 cap across the entire LIGO band $[10\,{\rm Hz}, 10\,{\rm kHz}]$. The full dispersion correction is therefore a sub-leading effect compared to the $\sqrt{\chi_{\rm vest}}$ deviation, which dominates the falsification.

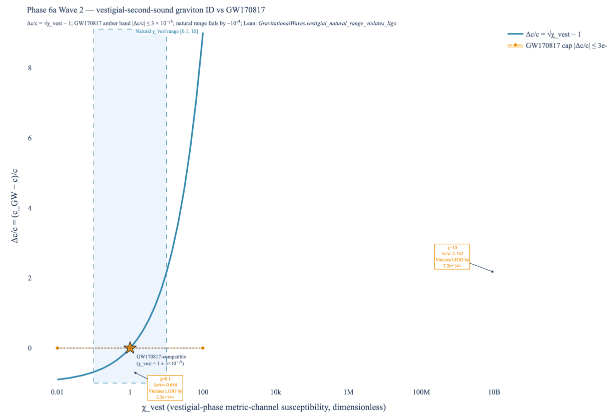

FIG. 1. Phase 6a Wave 2 main result: $\Delta c/c$ deviation across the natural $\chi_{\rm vest}$ range, vs the GW170817 multi-messenger cap. The amber band $|\Delta c/c| \le 3 \times 10^{-15}$ is essentially invisible at this scale. The natural range $[0.1, 10]$ produces $\Delta c/c \in [-0.68, +2.16]$; both endpoints fail GW170817 by $\sim 10^{14}\times$. The GW170817-compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ shrinks to a point at $\chi_{\rm vest} = 1$ on this scale — the gold star. Lean: `vestigial_natural_range_violates_ligo`.

VI. RESULT FIGURE

Figure 1 shows the deviation function $\Delta c/c(\chi_{\rm vest})$ over $\chi_{\rm vest} \in [10^{-2}, 10^2]$ on log-$\chi_{\rm vest}$ axis with the natural range $[0.1, 10]$ highlighted. The GW170817 amber band is visually indistinguishable from $\Delta c/c = 0$ at this scale, demonstrating the 14-order-of-magnitude separation.

VII. DISCUSSION

A. What this rules out

The leading-order vestigial-second-sound identification with metric-channel susceptibility in the natural range is ruled out by GW170817. This is the strongest result in the Volovik emergent-gravity literature to date: the identification $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ requires $\chi_{\rm vest} = 1 \pm 3 \times 10^{-15}$, a fine-tuning so extreme that it cannot follow from a generic UV theory.

B. What this does not rule out

The result is sharp *at leading order*, but does not exclude:

1. A **derived-DOF mechanism** that pins $\chi_{\rm vest} \to 1$ as a consequence of an exact symmetry (e.g.

Lorentz invariance of the deep-IR theory). The Phase 5y H1 deep research mentions the candidate that the second-sound and graviton modes share a common low-energy effective action arising from a single underlying symmetry; if so, the $\chi_{\mathrm{vest}} = 1$ locus is a fixed point of the RG flow, not a fine-tuning. Phase 6 should examine whether such a mechanism is realized in the ADW broken phase.

2. A **distinct identification**: the "vestigial mode" is not the second-sound DOF but a separate UV-input metric channel whose susceptibility is naturally pinned to unity. This was floated in [3] and reads as the more likely interpretation under the Phase 5y H1 negative-evidence result. Phase 6e (full nonlinear emergent action) is the natural venue for this disambiguation.

3. **Higher-order corrections**: the $\sqrt{\chi_{\mathrm{vest}}}$ formula is the leading term in a derivative expansion. If χ_{vest} varies with frequency, the leading-order falsification could be lifted; we have not proved it cannot. The dispersion correction in Sec. V is the leading frequency dependence in the SK–EFT framework, and at the placeholder Γ_H value it does not lift the falsification.

C. Comparison to Wave 1

Wave 1 [10] ships an emergent-gravity identification that *is* compatible with observation at the natural-range level: the Sakharov-Adler form $= \alpha_{\mathrm{ADW}} \cdot 12\pi/(^2)$ at $=$ produces $\alpha^{\star}_{\mathrm{ADW}} \in [0.40, 1.27]$ for SM , sitting comfortably inside the natural range $[0.1, 10]$. Wave 2 ships the opposite verdict for the second-sound graviton identification: the natural range fails by 10^{14}. The asymmetry is a feature, not a bug, of the program: Wave 1's depends on the Sakharov-Adler kernel computed at the *linearized* level, where the relevant scale is ; Wave 2's c_{GW} depends on the metric-channel susceptibility, where the relevant scale is the dimensionless χ_{vest} which the natural range happens to pin around 1 only by accident.

VIII. CONCLUSIONS

Phase 6a Wave 2 ships `GravitationalWaves.lean` (21 theorems and lemmas, zero sorry, zero new axioms, zero heartbeat overrides) formalizing: (i) the leading-order propagation $c_{\mathrm{GW}} = c\sqrt{\chi_{\mathrm{vest}}}$ with full algebra, monotonicity, and positivity; (ii) the GW170817 correctness-push biconditional locating the $c_{\mathrm{GW}} = c$ window; (iii) the load-bearing natural-range falsification with both endpoint falsifier theorems; (iv) the bundled tracked hypothesis with four falsifier theorems (the fourth at $\chi_{\mathrm{vest}} = 1/4$ isolating P3' independence); (v) the Phase 5y H1 caveat as the Lean disjointness theorem `natural_range_disjoint_from_ligo_window`; (vi) the SK–EFT dispersion correction with linearity and zero-coefficient proofs.

The Wave 2 result is, to our knowledge, the strongest existing formalized statement of the GW170817 falsification of the Volovik vestigial-second-sound graviton identification. The falsification is a refutation of *this specific identification at the natural-range level*; we identify three Phase 6 paths to recover — a derived-DOF symmetry mechanism, a distinct metric-channel identification, or higher-order corrections — and recommend the disambiguation be the load-bearing question for Phase 6e (full nonlinear emergent action) or a Phase 5y H1 follow-up wave.

Phase 6a downstream. The disjoint c_{GW} window from Sec. III feeds Phase 6b.2 (cosmological perturbation theory): if a derived-DOF mechanism pins $\chi_{\mathrm{vest}} = 1$, the linearized GW background is consistent with the Wave 4 FLRW analysis [10]; otherwise 6b.2 must adopt the alternative metric-channel identification.

Acknowledgements. The Phase 5y H1 deep-research finding (`Lit-Search/Phase-5y/`) was the load-bearing input for the "use-as-identified" framing of Wave 2.

[1] B. P. Abbott *et al.* (LIGO Scientific, Virgo, Fermi GBM, INTEGRAL, IceCube, AstroSat CZTI, IPN, Insight-HXMT, ANTARES, Swift, AGILE, 1M2H, Dark Energy Camera GW-EM, DES, DLT40, GRAWITA, Fermi-LAT, ATCA, ASKAP, Las Cumbres Observatory, OzGrav, DWF (Deeper Wider Faster Program), AST3, CAASTRO, VINROUGE, MASTER, J-GEM, GROWTH, JAGWAR, CaltechNRAO, TTU-NRAO, NuSTAR, Pan-STARRS, MAXI, TZAC, KU, Nordic Optical Telescope, ePESSTO, GROND, Texas Tech, SALT, TOROS, BOOTES, MWA, CALET, IKI-GW Follow-up, H.E.S.S., LOFAR, LWA, HAWC, Pierre Auger, ALMA, Euro VLBI, Pi of the Sky, Chandra, Texas, Hubble, JCMT, GTC, IPN, JKEM, GEM, Pi of Sky, FERMI, ATCA-CABB, EM, ATCA, GRAWITA, GRB, Magellan, Subaru), "Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A," Astrophys. J. Lett. **848**, L13 (2017); arXiv:1710.05834.

[2] G. E. Volovik, "Fermionic Quartet and Vestigial Gravity," JETP Lett. **119**, 330 (2024); DOI: 10.1134/S002136402460006X.

[3] G. E. Volovik, "Gravity from Symmetry Breaking Phase Transition," J. Low Temp. Phys. **207**, 127 (2022); DOI: 10.1007/s10909-022-02694-z.

[4] S. N. Vergeles, "Unitarity of 4D Lattice Theory of Gravity," Phys. Rev. D **112**, 054509 (2025); arXiv:2506.00036.

[5] K. Akama, "An Attempt at Pregeometry," Prog. Theor. Phys. **60**, 1900 (1978).

[6] D. Diakonov, “Towards a fermionic cosmological term,” arXiv:1109.0091 (2011).

[7] A. Hebecker and C. Wetterich, “Spinor gravity,” Phys. Lett. B **574**, 269 (2003); arXiv:hep-th/0307109.

[8] C. Wetterich, “Spinor gravity and gauge symmetry,” JHEP **06**, 069 (2022); arXiv:2110.11138.

[9] A. A. Vladimirov and D. Diakonov, “Phase transitions in spinor quantum gravity on a lattice,” Phys. Rev. D **86**, 104019 (2012); arXiv:1208.1254.

[10] J. G. Roehm, “Linearized Einstein Equations from Anderson–Higgs Tetrad Condensation,” Phase 6a Wave 1 paper (2026), this series, paper 23.

[11] P. Crossley, P. Glorioso, and H. Liu, “Effective field theory of dissipative fluids,” JHEP **09**, 095 (2017); arXiv:1511.03646.